

EMPORIA-GREENSVILLE, VA, USA

WMO: 720278

Lat: 36.687N

Lon: 77.483W

Elev: 127

StdP: 14.63

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03704

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.3	21.4	0.6	5.7	24.3	5.3	7.3	27.0	19.0	47.8	17.0	47.4	2.7	290	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	93.9	77.0	90.9	76.3	89.6	75.7	82.1	86.7	79.1	86.7	77.4	86.0	5.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.1	162.5	85.4	77.2	142.0	82.5	74.7	130.7	81.1	45.7	87.7	42.7	85.7	40.7	85.4	93.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	13.1	11.6	DB	9.0	98.0	7.6	3.4	3.5	100.5	-0.9	102.5	-5.2	104.4	-10.7	106.9	(4)
(5)			WB	7.7	82.7	7.3	4.2	2.5	85.7	-1.7	88.2	-5.8	90.5	-11.1	93.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.9	39.3	41.8	48.2	58.4	66.4	74.3	77.9	76.5	71.1	59.7	48.8	43.3		
	DBStd	15.77	10.26	9.63	9.82	8.41	7.39	5.48	4.28	4.68	5.71	8.31	8.23	9.16		
	HDD50	1184	357	266	157	20	0	0	0	0	0	13	119	252		
	HDD65	3669	796	652	525	230	78	4	0	1	207	487	674			
	CDD50	4432	26	36	103	273	507	728	865	820	633	313	84	44		
	CDD65	1438	0	1	5	32	120	282	400	356	198	42	1	1		
	CDH74	13159	0	8	74	409	1114	2675	3798	3165	1534	360	17	5		
	CDH80	5095	0	0	7	89	329	1063	1675	1326	524	82	0	0		
(14) Wind	WSAvg	4.1	4.6	5.0	5.5	5.5	4.2	3.6	3.3	3.0	3.6	3.8	4.0	3.9		
(15) Precipitation	PrecAvg	45.8	3.4	2.6	4.3	3.2	3.6	4.3	5.1	4.6	4.8	3.3	3.0	3.5		
	PrecMax	61.0	6.4	5.7	7.4	6.7	7.0	9.1	7.9	10.8	19.9	6.1	7.9	6.9		
	PrecMin	34.2	1.1	0.8	0.7	1.2	1.3	1.9	2.4	1.5	0.4	0.0	0.5	1.0		
	PrecStd	6.9	1.4	1.2	1.7	1.3	1.4	2.0	1.7	2.3	4.1	1.7	1.9	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.8	74.7	81.0	86.4	90.9	96.9	97.4	97.1	91.4	88.3	76.9	72.7		
		MCWB	62.5	64.1	64.8	67.4	74.2	77.1	77.7	79.1	74.4	72.0	64.4	66.1		
	2%	DB	66.2	69.9	75.3	82.0	86.5	91.1	93.4	92.7	89.6	81.4	72.0	68.1		
		MCWB	60.0	60.9	60.7	66.0	72.4	77.3	77.9	76.9	73.1	70.3	63.6	63.6		
	5%	DB	62.5	64.1	71.5	79.2	83.6	89.6	90.8	90.2	86.1	78.5	68.1	63.5		
		MCWB	56.1	57.0	59.3	64.5	71.7	76.4	76.9	76.0	72.4	68.0	61.1	59.4		
	10%	DB	56.5	59.4	66.1	74.6	80.9	86.3	88.4	87.8	82.1	73.3	63.6	59.3		
		MCWB	51.2	51.3	57.2	62.9	69.3	74.6	75.9	75.0	71.4	66.7	57.3	54.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.7	66.1	68.9	76.4	81.2	87.6	85.2	86.2	79.8	75.8	69.0	69.1		
		MCDB	69.4	71.4	73.0	81.0	83.2	89.6	90.7	91.5	84.2	82.0	71.7	70.1		
	2%	WB	62.0	62.8	64.3	70.2	74.7	80.7	81.5	80.4	76.8	73.4	65.3	63.6		
		MCDB	65.4	66.8	71.0	76.1	82.6	84.7	86.9	87.1	84.0	78.6	70.4	66.2		
	5%	WB	56.6	59.0	62.0	67.3	72.4	77.2	78.8	78.2	75.3	71.3	62.4	60.5		
		MCDB	60.9	64.3	68.4	75.7	80.4	86.8	87.5	86.1	82.1	76.1	65.8	63.1		
	10%	WB	51.8	52.6	58.8	64.5	70.7	75.4	77.2	76.6	73.7	68.4	59.1	55.4		
		MCDB	55.8	57.3	64.9	71.6	77.8	84.7	86.5	84.4	79.0	72.7	63.8	58.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	21.0	23.0	24.3	22.4	20.9	20.1	19.6	20.0	22.2	22.1	19.4		
		MCDBR	26.4	27.5	29.2	28.5	25.1	23.5	23.1	23.1	24.1	25.3	25.6	22.9		
		MCWBR	19.9	19.6	16.8	15.0	12.0	11.4	10.3	9.4	10.4	14.6	17.0	18.3		
	5% WB	MCDBR	23.9	25.4	24.6	23.0	21.9	21.4	20.6	19.7	18.8	21.0	20.3	20.9		
		MCWBR	20.3	20.8	17.4	15.7	12.9	12.8	11.1	9.8	10.0	13.6	15.7	17.9		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.316	0.347	0.390	0.436	0.487	0.514	0.512	0.404	0.357	0.321	0.304		
	taud		2.529	2.487	2.432	2.321	2.219	2.125	2.069	2.059	2.357	2.477	2.534	2.560		
	Ebn at Noon		285	293	292	283	270	256	247	245	269	274	275	277		
	Edh at Noon		25	29	34	40	45	49	52	51	36	29	25	23		
(44) All-Sky Solar Radiation	RadAvg		771	1000	1312	1668	1869	2017	1913	1732	1413	1126	854	677		
	RadStd		61	117	105	131	151	130	93	95	128	141	69	56		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (65 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	+288	+194

Nomenclature: See separate page